



Top predator absence enhances leaf breakdown in an intermittent stream



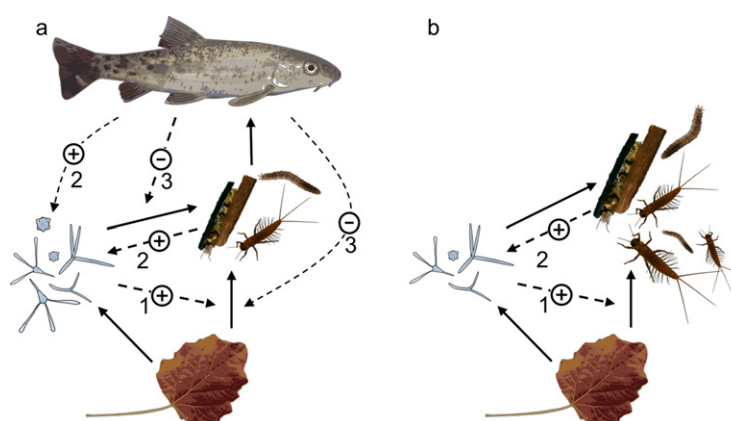
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HIGHLIGHTS

- Current Mass Extinction is characterized by the loss of top predators.
- There is a knowledge gap regarding the ecosystem consequences of fish loss in intermittent streams.
- We assessed if the presence/absence of an endangered fish affects leaf breakdown.
- Fish absence affected macroinvertebrate communities, leaf fungal biomass, and breakdown rates.
- Small-bodied fish can have a key role in ecosystem processes in intermittent streams.

GRAPHICAL ABSTRACT



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ABSTRACT

Current biodiversity loss is characterized by the extinction of top predators, but small-bodied freshwater fish are often overlooked in research and conservation management even when threatened because they usually lack commercial value. Therefore, the ecosystem impacts of their possible loss remain mostly unknown. We assessed whether the presence/absence of an endangered predatory fish (*Barbus meridionalis* (A. Risso, 1827)) in an intermittent stream affects leaf fungal biomass and leaf quality (i.e. leaf carbon:nitrogen ratio and leaf toughness), macroinvertebrate assemblages colonizing leaf packs, and leaf breakdown rates. We conducted a leaf bag experiment comparing a control reach with a population of *B. meridionalis* with an adjacent upstream fishless reach. In the fishless reach, leaf fungal biomass and microbially mediated breakdown rate were lower compared to the control reach. This was probably caused by the lack of the bottom-up stimulation through nutrient recycling by fish. Shredders and scrapers were found at higher abundance and biomass in the fishless compared to the control reach, and the whole macroinvertebrate community composition changed with fish absence. Consequently, macroinvertebrate mediated leaf breakdown was faster in the fishless than in the control reach, not only compensating for the lower microbially mediated leaf breakdown in the fishless reach, but accelerating the overall leaf breakdown rate. Our study contributes to understand the potential cascading effects produced by the extirpation of endangered small-bodied fish.

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1. Introduction

Our planet is experiencing the Sixth Mass Extinction, which is characterized by the loss of apex consumers (Duffy, 2002; Estes et al., 2011). Freshwater fish are among the most threatened fauna worldwide (Jenkins, 2003; Markovic et al., 2014), small-bodied ones being at greater risk of extinction than their larger-bodied counterparts (Olden et al., 2007). Despite their body size, small-bodied fish often act as top predators in several freshwater ecosystems, such as intermittent rivers, headwater permanent streams, and ponds (Meyer et al., 2007; Reich et al., 2010; Brucet et al., 2012). However, these species are often overlooked in research and conservation management even when considered threatened, probably because they usually lack recreational and commercial value (Braga et al., 2012; Saddlier et al., 2013) and inhabit ecosystems often neglected by conservation and management (Nikolaidis et al., 2013). Consequently, understanding the ecological consequences of the possible loss of small-bodied predatory fish is a current major challenge.

Trophic cascades have been documented mostly in grazing systems (see Estes et al., 2011), but many freshwater ecosystems, such as most forested headwater streams, are detritus-based systems where leaf-litter processing fuels food webs (Vannote et al., 1980; Tank et al., 2010). The hypothesis that detritus-based food webs are less likely to be top down-controlled compared with grazing systems (Rosenfeld, 2000) has been refuted by several studies that have demonstrated how predatory invertebrates can trigger trophic cascades in detritus-based streams (e.g. Lagrue et al., 2015). However, fish-induced trophic cascades in detritus-based streams are still scarcely found in the scientific literature (Greig and McIntosh, 2006; Buria et al., 2010) with some studies finding inconsistent (Ruetz et al., 2002) or no effects on leaf breakdown by predatory fish (Reice, 1991; Rosenfeld, 2000), raising doubts about the top-down control of leaf-litter processing by predatory fish.

Top predators can play a key role in nutrient recycling through the excretion and egestion of nutrients (Vanni, 2002; Schmitz et al., 2010). In freshwater systems, predatory fish can produce biogeochemical hotspots when they aggregate in local patches (McIntyre et al., 2008; Capps and Flecker, 2013). Although the potential effects of predatory fish in leaf breakdown through nutrient recycling are practically unexplored, several studies posit that nutrient enrichment in streams can increase leaf fungal biomass and, consequently, enhance microbially mediated leaf breakdown (Gulis and Suberkropp, 2003; Ferreira and Chauvet, 2011). Besides, higher nutrient availability in the streamwater can increase leaf N content, thus modifying leaf-litter quality (Gulis and Suberkropp, 2003). Therefore, given that fish extinctions can alter nutrient recycling in streams (McIntyre et al., 2007), fish absence may impact leaf-litter processing through bottom-up control.

In the present study, we assessed whether the presence/absence of an endangered predatory fish in an intermittent stream affects (1) leaf fungal biomass and leaf quality (i.e. leaf carbon:nitrogen (C:N) ratio and leaf toughness), (2) macroinvertebrate assemblages colonizing leaf packs, and (3) leaf breakdown rates. To assess the possible top-down impacts of small fish species we used, as a study case, the cyprinid *Barbus meridionalis* (A. Risso, 1827), a small-bodied fish listed as 'vulnerable' in the Spanish Red Book (Doadrio, 2001), 'near threatened' by the IUCN, and included in the EU's Habitats Directive and in the Bern Convention. We conducted a leaf bag experiment from late spring to early autumn in two adjacent reaches of an intermittent Mediterranean stream: a downstream reach with a population of *B. meridionalis* (control reach) and an upstream fishless reach. We hypothesized that macroinvertebrate communities are top-down controlled by *B. meridionalis*, as previous research has shown that this fish species can trigger a trophic-cascade on periphyton primary production (Rodríguez-Lozano et al., 2015b) and may feed preferentially on macroinvertebrate shredders (Rodríguez-Lozano et al., 2016). Therefore, the fishless reach would have higher shredder densities compared to the control reach and,

consequently, faster macroinvertebrate mediated litter breakdown rates (Fig. 1). Moreover, we hypothesized that *B. meridionalis* can create a biogeochemical hotspot in intermittent streams during the dry season when this species aggregates in pools, as do other fish species (Aparicio and de Sostoa, 1999). Hence, the absence of excretion and egestion by *B. meridionalis* may trigger a reduction in leaf fungal biomass, slowing down microbially mediated leaf breakdown.

2. Materials and methods

2.1. Study area

The study was carried out in the Castelló stream (lat 41°40'42"N, long 2°1'49"E; Altitude: 431 m a.s.l.) within the protected area of Sant Llorenç del Munt i l'Obac Natural Park (50 km inland from Barcelona city, NE Spain). This area is characterized by a calcareous geology and a Mediterranean climate, with mild winter and warm spring and summer. Rainfall is irregular and intense, occurring primarily in winter but also in spring and autumn, while summer is normally very dry. The protected area is dominated by Holm oak (*Quercus ilex* L.) and Aleppo pine (*Pinus halepensis* Miller) forests and Mediterranean shrubs (for a detailed description see Bonada et al., 2007). The Castelló stream is a first order stream, tributary of the Ripoll river in the Besòs basin, and its catchment is mainly forested, with small areas of cereal cultivation (<15%). The stream is formed by shallow riffles and pools; riffles usually dry during drought periods (i.e. commonly summer), while pools remain permanent acting as refuge for fish. We selected two adjacent 100-m reaches in the stream, separated by <0.5 km: a downstream reach with a population of *B. meridionalis* (control reach) and an upstream fishless reach. Both reaches were <2 m wide and <50 cm deep, had a bedrock stream bed, and were lined by a well developed riparian forest (>75% trees) with understory shrubs.

Other *Barbus meridionalis* populations were extirpated from near rivers and streams due to a wildfire in August 2003 and have not recovered since then possibly due to the presence of natural and human barriers (Rodríguez-Lozano et al., 2015b). The Castelló stream and its watershed were not affected by the wildfire, and its *Barbus meridionalis* population remains well preserved (Sostoa et al. 2006). The fish community in the downstream reach is only composed by the *Barbus meridionalis* population, which presents a low mean density of large individuals (e.g. 0.8 individuals m⁻² in 2005; Sostoa et al., 2006). However, fish aggregate in pools during drought periods, reaching densities >40 individuals m⁻² of populations dominated by young small fish (<4 cm fork length).

2.2. Field experiment

We conducted the leaf bag experiment from late spring to early autumn 2012 (14 June 2012–05 October 2012), coinciding with the peak of in-stream leaf inputs (Rodríguez-Lozano et al., 2015a) and with the aggregation of fish in pools. Leaves of white poplar (*Populus alba* L.) were collected just after abscission, and air-dried at room temperature to constant mass. Leaf bags (15 × 20 cm) containing 3 g of leaves (SE = 0.06 g) were made of 2 mesh sizes: coarse (10 mm) and fine (250 µm). Coarse-mesh bags allowed macroinvertebrate colonization and also small fish individuals to enter the leaf bags, thus more closely simulating natural leaf-litter breakdown processes, whereas fine-mesh bags excluded virtually all fish and macroinvertebrates, thereby allowing us to assess the relative contribution of microbial (fungi and bacteria) activity to leaf breakdown (Young et al., 2008).

We deployed 30 leaf bags of each mesh type in each reach (120 leaf bags in total), and we incubated an extra set of 12 additional bags in the fishless reach for 24 h to correct for initial leaf mass losses resulting from leaching and accidental transport losses (Gessner et al., 1999). We placed leaf bags in pools because leaves naturally accumulated in the pools of this stream, distributing them among three different pools at

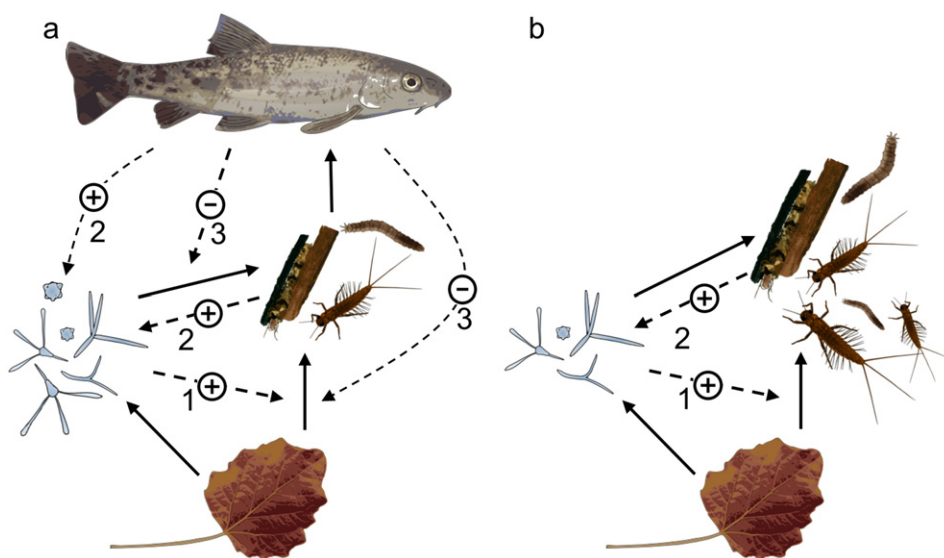


Fig. 1. Trophic (solid) and non-trophic (dashed) interactions among the predatory fish, shredders, fungal decomposers, and leaf-litter in a) the presence and b) absence of the top predator. Non-trophic interactions include: (1) fungal facilitation of shredders through leaf conditioning, (2) fungal stimulation through nutrient excretion by fish and invertebrates, and (3) trait-mediated reduction of foraging activity. Adapted from Jabiol et al. (2013).

each reach to better represent the spatial variability of the reaches. We removed 6 litter bags of each type (fine- and coarse-mesh) from each reach 5, 12, 26, 58, and 113 days after deployment. Leaf bags were carefully retrieved, placed individually in zip-lock bags with streamwater, and transported in refrigerated containers to the laboratory, where we processed them immediately. We washed the material in each litter bag, collected invertebrates on a 250- μ m sieve, and preserved them in 70% ethanol. For each litter bag, leaf discs of 10-mm diameter were cut and stored to quantify leaf toughness, leaf C:N ratio, and leaf fungal biomass. The remaining leaf material was oven-dried (60 °C, 72 h) and combusted (500 °C, 12 h) to measure remaining ash-free dry mass (AFDM).

2.3. Physicochemical measurements

We carried out physicochemical measurements throughout the litter breakdown experiment ($n = 6$). We measured dissolved O_2 , pH, and conductivity in situ with a Multiline P4 WTW meter (YSI, Yellow Springs, OH, U.S.A.), and we estimated discharge from mean depth, transect width, and water velocity with a flow meter (miniAir, Schiltknecht, Gossau, Switzerland). We collected water samples by plastic bottles 10–30 cm below the water surface, reserved 50 mL for total organic carbon analysis (TOC), and filtered the rest of the water through GF/F Whatman filters to analyze dissolved organic carbon (DOC) and nutrients. TOC and DOC were determined in an automated N, C analyzer (multi N/C 3100, Analytik Jena, Jena, Germany). We measured ammonium ($NH_4^+ - N$) and soluble reactive phosphorous ($PO_4^{3-} - P$) with spectrophotometric methods (Shimadzu UV-1201, Shimadzu, Tokyo, Japan) and nitrate ($NO_3^- - N$) with ionic chromatography methods (UV/V Kontron model 332, Kontron AG, Zürich, Switzerland). Water temperature was recorded every 10 min throughout the leaf bag experiment using submersible temperature data loggers (HOBO Pendant, Onset Computer Corporation, Bourne, MA, U.S.A.).

2.4. Laboratory analysis

Fungal biomass in leaf-litter was assessed through determination of ergosterol content, following Gessner (2005). Five frozen leaf discs of each treatment replicate ($n = 3$) were freeze-dried, weighed to the nearest 0.1 mg, and heated at 80 °C for 30 min with alkaline methanol to extract the lipids. Subsequently, extracts were purified using solid-phase extraction cartridges (Sep-Pak tC18, 500 mg, 3 cm³, Waters

Corporation, Milford, MA, U.S.A.). Ergosterol concentration was quantified by comparing absorbance at 282 nm after separation from other lipids by high-performance liquid chromatography (HPLC, JASCO, Easton, MD, U.S.A.; Column Gemini® 5 μ m NX-C18 250 $\times$ 4.6 mm, Phenomenex, Torrance, CA, U.S.A.) with a standard of ergosterol (Sigma-Aldrich, St. Louis, MO, U.S.A.). Ergosterol content was converted to mycelium biomass using the average conversion factor of 5.5 mg of ergosterol per gram of mycelium (Gessner and Chauvet, 1993). We standardized fungal biomass per gram of leaf-litter AFDM remaining in leaf bags. Leaf toughness was estimated by measuring the force needed to penetrate leaves using a penetrometer (punch diameter = 1.95 mm), following methods described by (Graça and Zimmer, 2005). For each treatment replicate ($n = 6$), leaf toughness corresponded to the mean of 3 measurements on 3 distinct leaves. To determine the C:N ratio we grounded together 3 leaf discs of 3 distinct leaves for each treatment replicate ($n = 3$). Subsequently, leaf C and N contents were quantified using a gas chromatograph coupled to a TCD detector after combustion at 1000 °C.

We counted macroinvertebrates, identified them to the lowest possible taxonomic level (usually genus), and measured them to the nearest mm. We classified taxa to functional feeding groups (FFG) following Tachet et al. (2010). We calculated individual biomass using published body length–dry mass equations (Dumont et al., 1975; Smock, 1980; Meyer, 1989; Benke et al., 1999; Miserendino, 2001; Baumgärtner and Rothhaupt, 2003; Ohta et al., 2011), except Hydracarina, Ostracoda, Cladocera, Oligochaeta, Nematoda, and Tardigrada for which we used biovolume data (Ramsay et al., 1997). We standardized macroinvertebrate abundance and biomass in coarse-mesh leaf bags per gram of remaining AFDM of leaf-litter in leaf bags. We also calculated macroinvertebrate mean body size (as body length) of each coarse-mesh litter bag.

2.5. Data analysis

We used an exponential decay model (Bärlocher, 2005) to quantify leaf breakdown rates:

$$M_t = M_0 e^{-kt}$$

where M_t is the leaf-litter AFDM at time t , M_0 is the initial AFDM corrected for leaching and transportation mass losses, $-k$ is the

breakdown rate, and t is the time in days. To test for significant differences in leaf breakdown rates among sites and mesh sizes, we conducted an analysis of covariance (ANCOVA) on $\ln(x)$ -transformed AFDM remaining (dependent variable), with site and mesh size as fixed factors and time (d) as a covariate (Zar, 2010). We corrected for temperature effects on leaf breakdown rates, by repeating the analyses using degree days (dd) instead of days as the covariate. We calculated degree days as mean daily temperatures accumulated by each sampling day (Menéndez et al., 2003). Similarly, to detect differences in leaf toughness, we conducted ANCOVAs with site and mesh size as fixed factors and time as covariate. To test for significant differences in leaf fungal biomass and leaf C:N ratio, we conducted three-way analysis of variance (ANOVAs) with site, mesh, and time as independent variables.

Nonmetric multidimensional scaling (NMDS) ordination was employed to examine the similarity in the macroinvertebrate composition of the coarse-mesh bags. To test for differences in the macroinvertebrate community composition between reaches, we used permutational multivariate analysis of variance (PERMANOVA, 'Adonis' function in R) on the Bray-Curtis distance matrix, after the $\log(x)$ -transformation of the macroinvertebrate biomass data. Subsequently, we used indicator species analysis, using 'IndVal' test in R, to identify which macroinvertebrate taxa of the coarse-mesh bags could serve as indicator for *B. meridionalis* presence or absence. The 'IndVal' test calculated the indicator value for each taxon, combining measurements of taxon specificity to each reach with taxon fidelity within each reach (Dufrene and Legendre, 1997). The significance of 'IndVal' measures was tested using the Monte Carlo test with 9999 permutations. To test for significant differences in total macroinvertebrate abundance and biomass in coarse-mesh leaf bags between reaches, we performed two-way analysis of variance (ANOVAs) with site and time as independent variables. We did the same analysis for each macroinvertebrate FFG, applying Bonferroni corrections to control for comparison-wise error. We also tested for differences between reaches in mean body size of macroinvertebrates in coarse-mesh bags. All statistical analyses were performed using the programs R (version 2.15.2, R Project for Statistical Computing, Vienna, Austria).

3. Results

Both reaches had alkaline pH, low discharge, and low nutrient concentrations during the study period (Table 1). Reaches only differed in dissolved oxygen levels, which were slightly higher in the fishless than in the control reach (Fish, $F_{1,16} = 21.65$, $P < 0.001$; Table 1). In the six field visits, pools were the dominant mesohabitat but still connected by thin water threads (flow $< 1 \text{ L s}^{-1}$). Flow was higher during late spring (experimental days 0, 5, and 12) and early autumn (experimental day 113), while was minimum during summer (experimental days 26 and 58) when riffles were practically inexistent ($< 10 \text{ cm}$ wide, $< 2 \text{ cm}$ deep). Therefore, although pools were not observed completely disconnected, most probably pools were disconnected, and reconnected by following rains, at some point during the last 87 days

Table 1
Mean (± 1 SE) values of characteristics of the control and the fishless reaches during the litter breakdown experiment (14 June 2012–05 October 2012).

	Control	Fishless
Discharge (L s^{-1})	0.7 ± 0.1	0.6 ± 0.2
Water temperature ($^{\circ}\text{C}$)	6.82 ± 0.05	6.60 ± 0.04
pH	8.16 ± 0.06	8.22 ± 0.05
Conductivity ($\mu\text{S cm}^{-1}$)	478 ± 5	474 ± 20
DO (% saturation)	85.7 ± 0.9	95.1 ± 0.1
$\text{NH}_4^+ - \text{N}$ ($\mu\text{g L}^{-1}$)	50 ± 7	48 ± 1
$\text{NO}_3^- - \text{N}$ ($\mu\text{g L}^{-1}$)	102 ± 16	97 ± 27
$\text{PO}_4^{3-} - \text{P}$ ($\mu\text{g L}^{-1}$)	22 ± 4	28 ± 1
TOC (mg L^{-1})	3.59 ± 0.21	3.60 ± 0.24
DOC (mg L^{-1})	3.06 ± 0.11	3.16 ± 0.18

of the experiment. During the experiment, fish were observed foraging near and inside the leaf bags.

Poplar-leaf mass loss fit an exponential model with a similar adjust for days and degree days (Table 2). Leaves in the coarse-mesh bags at the fishless reach presented the fastest breakdown rate, and leaves in the fine-mesh bags at the same reach the lowest rate. Litter mass loss over time (i.e. without temperature correction) differed between the fishless and the control reach (Fish * Time, $F_{1,112} = 13.06$, $P < 0.001$; Fig. 2a). Litter breakdown rate also differed between mesh sizes (Mesh * Time, $F_{1,112} = 73.68$, $P < 0.001$; Fig. 2a), and a significant interaction among factors was detected (Fish * Mesh * Time, $F_{1,112} = 38.65$, $P < 0.001$). When using accumulated heat (degree days) instead of time (days) as independent variable (Fig. 2b), we obtained the same results (Fish * Degreedays, $F_{1,112} = 16.12$, $P < 0.001$; Mesh * Degreedays, $F_{1,112} = 74.43$, $P < 0.001$; Fish * Mesh * Degreedays, $F_{1,112} = 40.25$, $P < 0.001$). Pairwise comparisons showed that litter breakdown in coarse-mesh bags was significantly faster in the fishless than in the control reach (Fish * Time, $F_{1,56} = 44.71$, $P < 0.001$), whereas litter breakdown in fine-mesh bags was marginally faster in the control compared to the fishless reach (Fish * Time, $F_{1,56} = 3.69$, $P = 0.06$). Litter breakdown for coarse-mesh bags was faster compared to fine-mesh bags only in the fishless reach (Mesh * Time, control: $F_{1,56} = 2.46$, $P = 0.12$; fishless: $F_{1,56} = 126.8$, $P < 0.001$).

Poplar leaf toughness (Fig. 3a) decreased over time (Time, $F_{4,100} = 244.75$, $P < 0.001$). Leaf toughness was lower in the fishless than in the control reach (Fish, $F_{1,100} = 11.11$, $P = 0.001$) and in coarse- than fine-mesh leaf bags (Mesh, $F_{1,100} = 5.11$, $P = 0.026$). Leaf fungal biomass (Fig. 3b) was higher in the control than in the fishless reach (Fish, $F_{1,40} = 34.72$, $P < 0.001$) and in coarse- than fine-mesh bags (Mesh, $F_{1,40} = 27.41$, $P < 0.001$). The temporal pattern in leaf fungal biomass differed between the fishless and the control reach (Fish * Time, $F_{4,40} = 29.99$, $P < 0.001$). Leaf C:N ratio (Fig. 3c) decreased over time (Time, $F_{4,40} = 15.40$, $P < 0.001$), but no effect of *B. meridionalis* presence nor mesh size was detected ($P > 0.6$).

The NMDS plot based on biomass data of macroinvertebrate taxa provided an interpretable two-dimensional ordination of the macroinvertebrate assemblages of the coarse-mesh leaf bags (Fig. 4). The taxonomic composition of the macroinvertebrate assemblages differed significantly depending on *B. meridionalis* presence (Fish, $F_{1,50} = 30.86$, $P < 0.001$). Macroinvertebrate composition also changed over time (Time, $F_{4,50} = 10.03$, $P < 0.001$), with a significant interaction between factors (Fish * Time, $F_{4,50} = 5.76$, $P < 0.001$): the differences in macroinvertebrate assemblages between reaches decreased during the leaf breakdown process (Fig. 4). Eight taxa were identified as indicators of fish presence and six taxa of fish absence (Table 3). The shredders *Habrophlebia* sp. and Stratiomyidae, as well as several scrapers (i.e. *Gyraulus* sp., *Radix* sp., and *Ancylus* sp.), were characteristic of the leaf bags at the fishless reach. The shredder *Thraulus bellus* and the Orthocladinae family (scrapers) were characteristic of *B. meridionalis* presence. Other shredders, such as *Mystacides* sp., Limnephilidae, and Tipulidae were collected in leaf bags, but these were not indicators of top predator presence or absence and had low abundances.

Total macroinvertebrate abundance in the coarse-mesh bags did not differ between reaches (Fish, $F_{1,50} = 1.23$, $P = 0.27$), while total macroinvertebrate biomass (Fig. 5a) was higher in the fishless reach

Table 2
Mean (SE) leaf-litter breakdown rates ($-k$) and R^2 values for regressions of $\ln(x)$ -transformed leaf mass remaining vs time in days (d) and cumulative degree days (dd).

Site	Mesh	d			dd		
		$-k$	SE	R^2	$-k$	SE	R^2
Control	Coarse	0.00786	0.00036	0.94	0.00115	0.000050	0.95
Control	Fine	0.00710	0.00033	0.94	0.00104	0.000047	0.95
Fishless	Coarse	0.01100	0.00031	0.98	0.00164	0.000047	0.98
Fishless	Fine	0.00627	0.00029	0.95	0.00093	0.000043	0.94

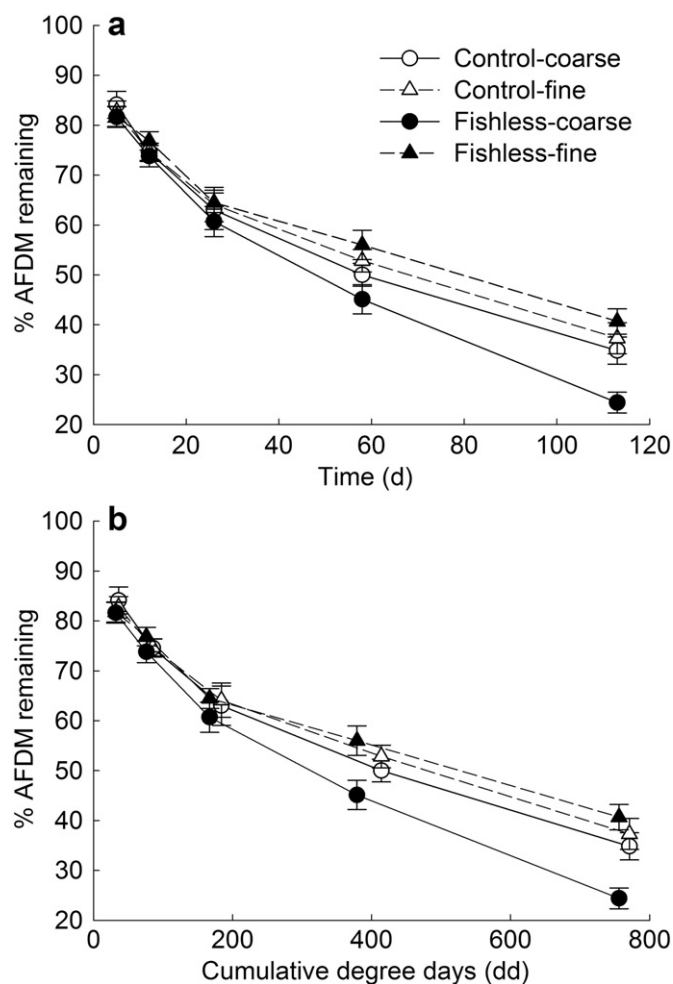


Fig. 2. Mean (± 1 SE; $n = 6$) % initial *Populus alba* leaf-litter mass remaining in coarse- and fine-mesh bags at the control and the fishless reaches over 113 days expressed a) over time and b) over cumulative degree days.

compared to the control reach (Fish, $F_{1,50} = 6.33$, $P = 0.015$). Mean body size was higher in the coarse-mesh leaf bags at the fishless than at the control reach (Fish, $F_{1,50} = 8.51$, $P = 0.005$). In terms of functional feeding groups (FFG), the abundance and biomass of shredders and scrapers (Fig. 5b–c) was higher in the coarse-mesh bags of the fishless reach compared to the control reach, whereas the abundance and biomass of gathering and filtering collectors (Fig. 5e–f) was greater in the control than in the fishless reach ($F_{1,50} = 7–64$, $P = 0.01$ to <0.001). Abundance and biomass of predatory invertebrates in coarse-mesh bags did not differ between reaches ($P > 0.9$; Fig. 5d).

4. Discussion

4.1. Microbially mediated leaf breakdown

Leaf fungal biomass was bottom-up stimulated through nutrient recycling by the top predator and macroinvertebrates. Fish absence negatively affected leaf fungal biomass and, consequently, microbially mediated leaf breakdown was slower in the fishless reach compared to the control one (i.e. marginally significant differences between reaches in leaf breakdown in fine-mesh bags). We could hypothesize that the lower leaf fungal biomass at the fishless reach compared to the control reach was caused by the higher density and biomass of scrapers and shredders (i.e. that scrapers and shredders top-down controlled leaf fungal biomass). However, leaf fungal biomass in the fine-mesh bags, into which macroinvertebrates had no access, was lower in the fishless

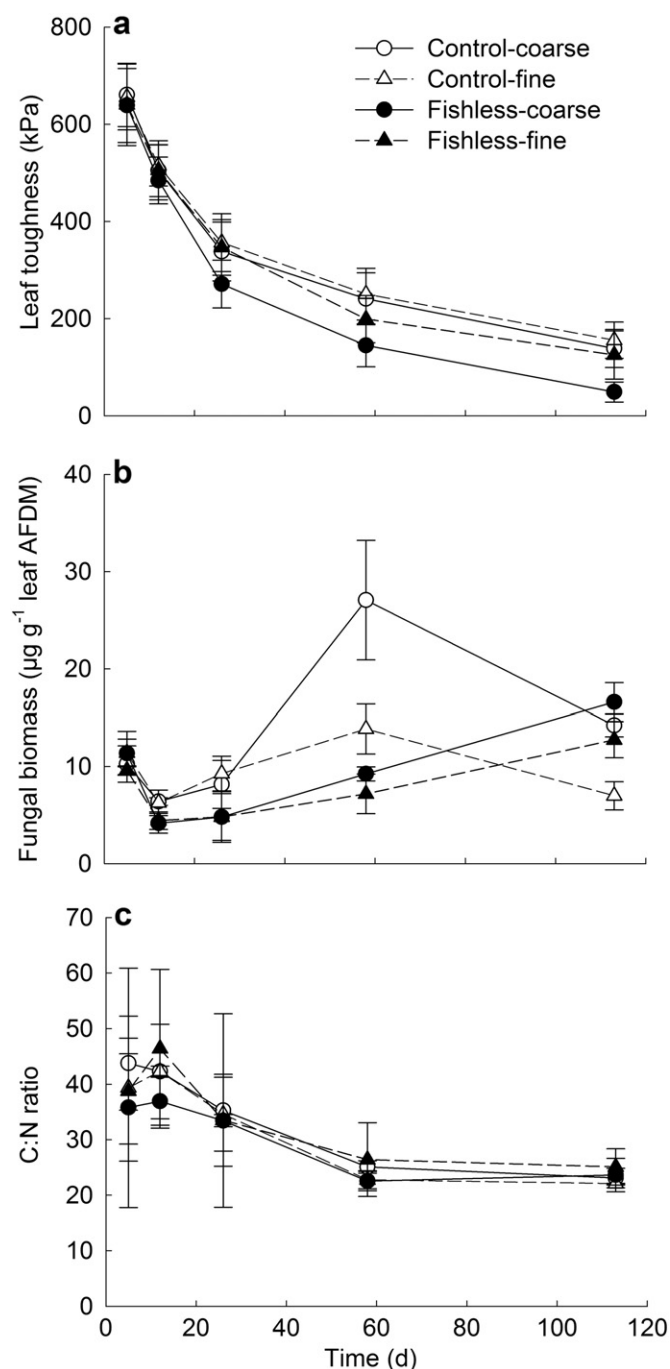


Fig. 3. Mean (± 1 SE) a) leaf toughness ($n = 6$), b) leaf fungal biomass ($n = 3$), and c) C:N ratio ($n = 3$) in coarse- and fine-mesh bags at the control and the fishless reaches over the experimental period.

than in the control reach, thus leaf fungal biomass was negatively affected by the absence of *B. meridionalis* independently of the density and biomass of macroinvertebrates. Moreover, fungal biomass was higher in coarse-mesh compared to fine-mesh bags, indicating that macroinvertebrates positively affected fungal biomass rather than negatively through top-down control. Most likely, macroinvertebrates egestion and excretion improved the micro-environmental conditions for fungal growth in coarse-mesh leaf bags. In a lacustrine environment, fish presence increased leaf fungal biomass in summer, but not in winter when fish density was lower (Mancinelli et al., 2007). In our experiment, fish aggregated in pools at high densities which may have created biogeochemical hotspots (McIntyre et al., 2008; Capps and Flecker, 2013).

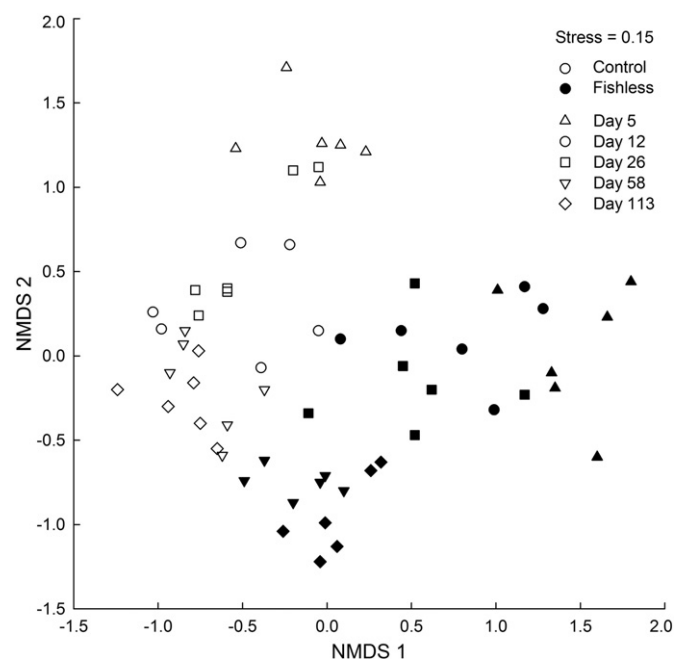


Fig. 4. Nonmetric multidimensional scaling (NMDS) ordination of macroinvertebrate assemblages in coarse-mesh bags. Colors distinguish *Barbus meridionalis* presence (white) and absence (black) and different symbol shapes denote different sampling dates.

Because the experiment was carried out in summer, water temperature (mean daily temperature was 6.71 °C with maximums >10 °C) could have had a synergistic effect with nutrient recycling over leaf fungal biomass (Ferreira and Chauvet, 2011). In general, the role of predators in the microbial communities involved in leaf-litter processing in freshwater systems is highly unknown (but see Majdi et al., 2015). The present study is the first to have specifically explored the impact of a predatory fish on leaf fungal biomass in a stream.

Although *B. meridionalis* presence increased leaf fungal biomass, it had no impact on streamwater TOC, DOC, and nutrient concentrations. We hypothesize that top predator impact on nutrient recycling was produced through excretion of metabolic waste, thus labile carbon and nutrients entered in the fast cycle in inorganic forms and were rapidly taken up by the microbial assemblages, moving directly to the stream food web (Schmitz et al., 2010; Atkinson et al., 2014). Although we only measured fungal biomass, bacteria most likely had an important role in this process too. Microbial assemblages (i.e. both fungi and bacteria) in leaf-litter can obtain nutrients from the leaves and from the streamwater. The proportion of streamwater N and C that is used by

microbial assemblages in leaf-litter increases during the breakdown process (Pastor et al., 2014), and in the case of N can reach >80% (Cheever et al., 2013). Moreover, our study site was an oligotrophic stream and *Populus alba* leaves were initially not rich in N ($1.3 \pm 0.1\%$), thus microbial decomposers were probably limited by nutrients and labile C, being more easily influenced by fish and macroinvertebrate effects on nutrient recycling (Ferreira et al., 2006; Vanni, 2010). In fact, the LINX II project (Mulholland et al., 2008) demonstrated that the efficiency of N biotic uptake from streamwater increased as N concentration in water column decreased.

Leaf C:N ratio decreased over time, most likely because of microbial immobilization of nitrogen (Ferreira et al., 2006). *Barbus meridionalis* had no impact on leaf C:N ratio despite its positive effect on leaf fungal biomass. Previous research has shown that nutrient enrichment may increase leaf fungal biomass, while having no impact on N leaf content (Ferreira et al., 2006). Robinson and Gessner (2000) showed that nutrient addition increased N and P in leaves, but the effect disappeared after three weeks. Initial leaf toughness can be a significant predictor of leaf breakdown rates and leaf consumption by invertebrates and microorganisms when comparing different leaf species (Graça and Zimmer, 2005). In our study, leaf toughness followed a similar trend to that observed for leaf mass loss and thus seems to be a suitable alternative measure for assessing leaf breakdown (Young et al., 2008).

4.2. Macroinvertebrate mediated leaf breakdown

Overall leaf litter breakdown was faster in the fishless than in the control reach and, therefore, macroinvertebrates accelerated leaf-litter breakdown in the fishless reach more than compensating for the slower microbially mediated leaf breakdown. We dismissed the effects of possible differences in current velocity, mechanical fragmentation, and sediment deposition on leaf decomposition between bags with different mesh sizes. Physical fragmentation and abrasion of leaves were unlikely to be important for leaf breakdown in our study because leaves were deployed in pools at water velocity < 0.15 m s⁻¹, far below threshold velocities reported to affect leaf-breakdown rates (Ferreira et al., 2006). Moreover, we saw no evidence of the mechanical fragmentation of leaves. Sediment deposition was unlikely to be important to leaf breakdown because no differences in substrata characteristics were discernible between sites, and burial of leaf bags by sediment was not observed. We isolated the effects of macroinvertebrates on leaf breakdown rates by subtracting *k*-values in fine-mesh bags from those in coarse-mesh bags. These calculations showed that the macroinvertebrate mediated leaf breakdown rate was 6.2 times higher in the fishless than in the control reach (i.e. 0.00473 vs 0.00076 when expressed per day). This difference was probably caused, at least partially, by intersite differences in shredder density and biomass in coarse-mesh bags, which were significantly higher in the fishless than in the control site. Scraper density and biomass were also higher in coarse-mesh bags in the fishless stream, which could also have contributed toward increasing leaf breakdown since scrapers, such as gastropods, can also have a positive effect on leaf breakdown rates (Treplin and Zimmer, 2012).

The macroinvertebrate contribution to leaf breakdown was negligible in the control reach (i.e. no significant differences in breakdown rates between fine- and coarse-mesh bags), although shredders and scrapers were present in coarse-mesh bags. This suggests that *B. meridionalis* presence not only had a density-mediated impact, but also a trait-mediated, reducing macroinvertebrate feeding activity on leaf-litter even despite of the higher leaf palatability (i.e. higher leaf fungal biomass) in the control compared to the fishless reach. Schmitz et al. (2004) suggested that trait-mediated responses of prey to predators ultimately determine trophic cascades, but studies on behaviorally mediated effects of predatory fish on detritus-based systems are scarce and their results are equivocal (Greig and McIntosh, 2006; Boyero et al.,

Table 3

Macroinvertebrate taxa detected as significant indicators for the coarse-mesh leaf bags in the control and the fishless reaches. T – Treatments: C = control, F = fishless. IndVal – indicator value. *P* – its respective *p*-value.

Taxa	T	IndVal	<i>P</i>
Cladocera	C	80.17	<0.001
<i>Caenis</i> sp.	C	73.44	<0.001
Oligochaeta	C	62.00	0.003
<i>Corynoneura</i> sp.	C	60.17	0.010
Orthocladinae	C	59.00	0.002
Ceratopogoninae	C	51.82	0.003
<i>Thraulius bellus</i>	C	23.33	0.011
<i>Planaria torva</i>	C	20.00	0.024
<i>Radix</i> sp.	F	79.85	<0.001
<i>Gyraulus</i> sp.	F	72.89	<0.001
<i>Habrophlebia</i> sp.	F	56.16	<0.001
Copepoda	F	46.56	0.008
Stratiomyidae	F	29.95	0.003
<i>Ancylus fluviatilis</i>	F	26.67	0.005

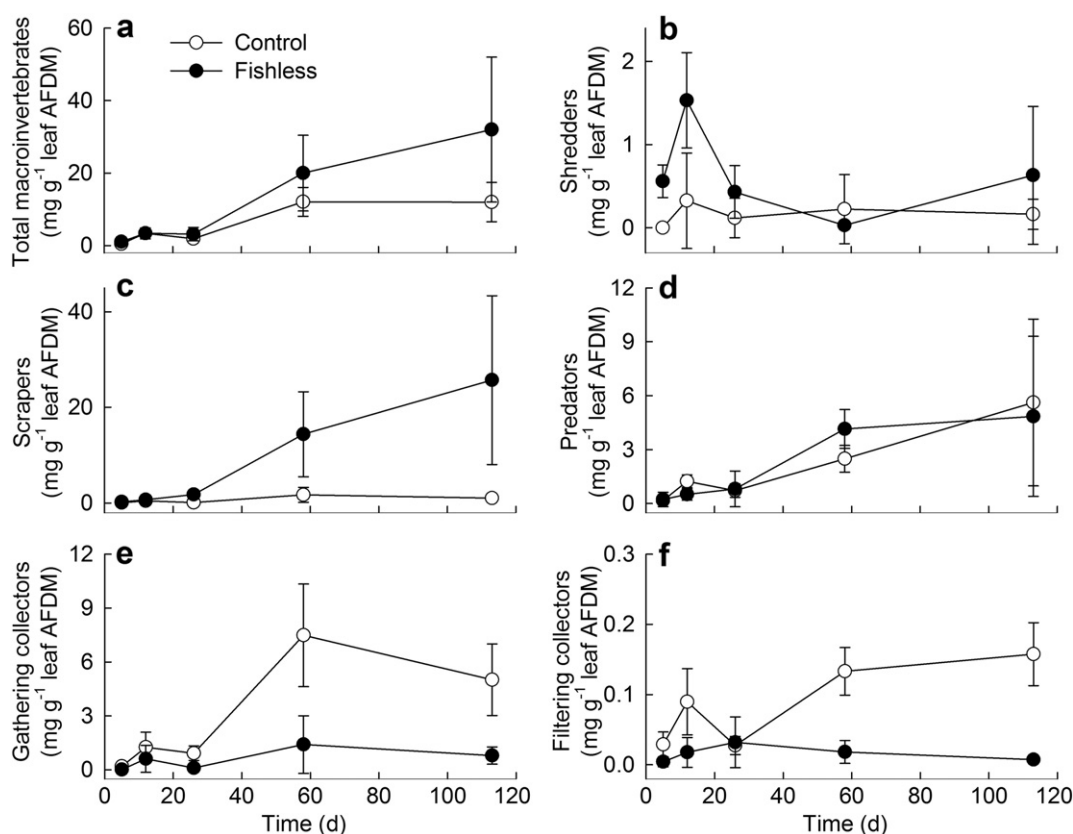


Fig. 5. Mean (± 1 SE; $n = 6$) biomass (mg g^{-1} remaining leaf ash-free dry mass [AFDM]) of a) total invertebrates, b) shredders, c) scrapers, d) predators, e) gathering collectors, and f) filtering collectors in coarse-mesh bags at the control and the fishless reaches over the experimental period.

2008; Jabiol et al., 2013). Several laboratory studies have found that shredders may reduce their activity in presence of fish chemical cues (e.g. Boyero et al., 2008), but only in some cases it led to a reduction in leaf breakdown (Short and Holomuzki, 1992; Rezende et al., 2015). The trait-mediated response of prey could depend on the predator-prey pair as a function of the prey vulnerability to the predator and the existence of other antipredator strategies (Schmitz et al., 2004; Jabiol et al., 2013).

Moreover, *B. meridionalis* did not only reduce some taxa density, but it changed the whole macroinvertebrate assemblage associated to leaf bags. For instance, *B. meridionalis* favored filtering collectors, such as cladocerans, which may be caused by the fish foraging movements that can dislodge and resuspend deposited sediments (Gelwick et al., 1997). The capacity of *B. meridionalis* to change the composition of the whole macroinvertebrate community has been reported before (Rodríguez-Lozano et al., 2015b). The lower macroinvertebrate sizes in top predator presence compared to its absence was probably caused by the observed changes in community composition (i.e. several small invertebrate taxa were favored, such as cladocerans and Orthocladinae) and by the size-selectivity feeding of *B. meridionalis* (Rodríguez-Lozano et al., 2016). In our study, the differences in the assemblage composition due to the top predator absence could have affected leaf breakdown rates if the favored taxa did not functionally compensate the reduction or loss of other taxa (Ernest and Brown, 2001; Rodríguez-Lozano et al., 2015b). This also constitutes an alternative explanation to the negligible contribution of macroinvertebrates to leaf breakdown in the control reach. Greig and McIntosh (2006) found no differences in total shredder biomass among streams with and without brown trout, but higher leaf breakdown rates in fishless streams; it was explained by changes in the composition of the shredder assemblages as an efficient obligate shredder was only present in fishless streams. This suggests that the impact of a predatory fish on leaf breakdown

can depend on the interaction strength between shredders and leaf-litter, shredder identity being a key factor.

4.3. Conclusions

The present study is the first to have assessed a trophic cascade of a predatory fish over leaf breakdown in an intermittent stream. Our results suggest that the extirpation of an endangered small-bodied fish may lead to substantial ecosystem changes in a detritus-based stream, simplifying the functional complexity of the system. Top predator absence was associated to a lower leaf fungal biomass and, consequently, microbially mediated breakdown was slower in the fishless than in the control reach. This was probably caused by the lack of the bottom-up stimulation through nutrient recycling by fish. Fish absence also changed the composition of macroinvertebrate assemblages, increased invertebrate mean body size, led to higher shredder and scraper abundances, and most likely enhanced the foraging activity of some invertebrates, which contributed significantly to the trophic cascade. Consequently, macroinvertebrate mediated leaf breakdown was faster in the fishless than in the control reach, not only compensating for the lower microbially mediated leaf breakdown, but accelerating the overall leaf breakdown rate. The key role of small-bodied fish in stream food-webs may increase during the dry season due to fish aggregation in pools, particularly in intermittent streams. This is highly relevant because, although intermittent streams have been neglected by research and conservation (Nikolaidis et al., 2013), these systems comprise approximately one half of the global river network and are projected to increase due to climate change and water abstraction for human use (Carlisle et al., 2011). Given the high extinction risk of small-bodied freshwater fish species, we think that conservation efforts must be undertaken due to their inherent value and the major ecosystem changes that may be caused by their loss.

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